

"MATLAB HACKTHON: MICROGRID ON MARS"

Submitted by:

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TASK 1 - DESIGN SPACE EXPLORATION

In the first task, the design was performed using the Simulink model *Task1 ideal/System* to assess how varying solar panel ratings and battery state-of-charge (SOC) levels influence the microgrid's ability to sustain uninterrupted operation for a 7-sol Martian period.

Simulations were carried out for solar power ratings in increments of 10 kW, ranging from 10 kW to 60 kW, and initial battery SOC in steps of 10%, ranging from 20% to 100%. After iterative runs, the optimal operating point was found at:

- **Solar Panel Rating:** 50kW
- **Initial Battery SOC:** 90%

This configuration ensured that the battery did not deplete or overcharge while meeting power demand. The resulting energy balance provided a feasible baseline for the subsequent hardware design and control tasks.

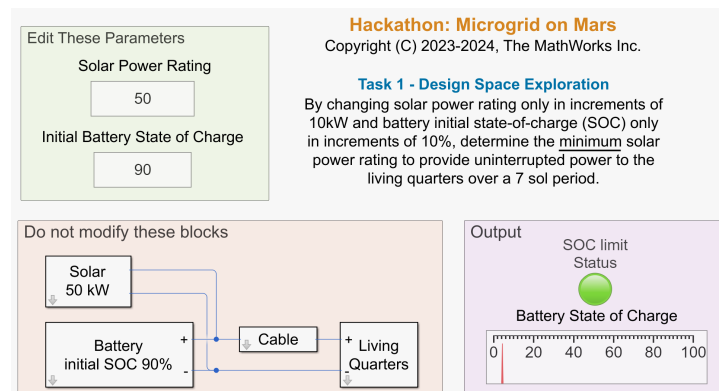


Image Task1_1

TASK 2 - DETERMINE MAXIMUM POWER TRANSFER

The goal was to identify the point of maximum power transfer from the solar panel using the principle of impedance matching. The model was configured to analyze the effect of varying load resistance on solar output power. A parametric sweep of resistive load values was conducted to locate the resistance value at which the output power peaked.

The maximum power was observed at a load resistance of **5.07 Ohms**, resulting in an output

power of approximately **50 kW**.

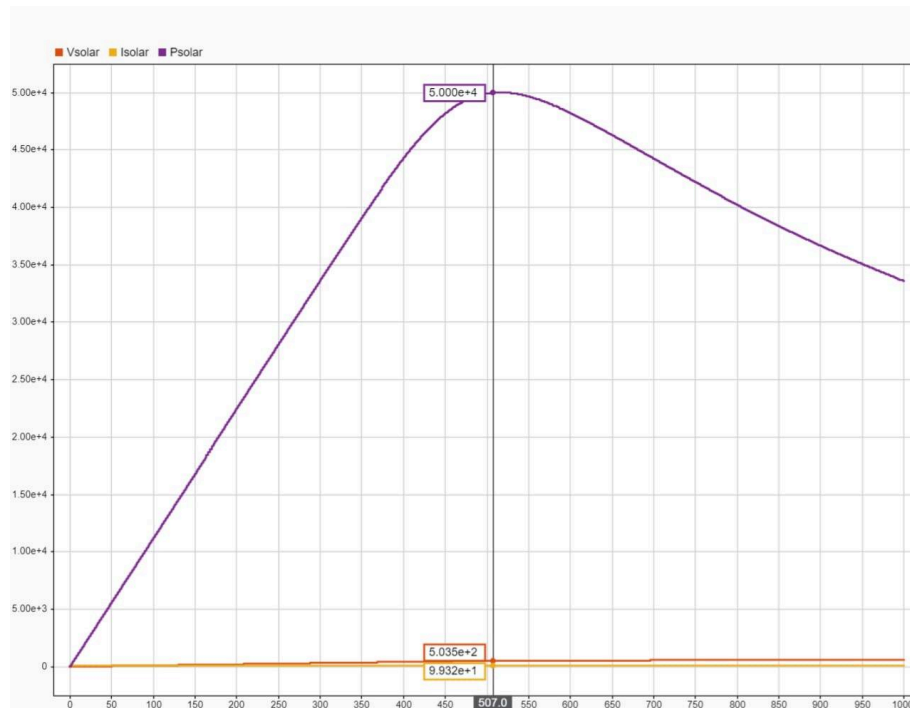


Image Task2_1

A scaled array was designed using this series-parallel configuration:

- Number of Panels in Series (N_s): 1000
- Number of Strings in Parallel (N_p): 10

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Task 2 - Determine Maximum Power Transfer

Replace the Load block with a variable resistor block and sweep through the resistance value from 10 to 1 Ohms. Then, modify the settings of the Solar Cell to scale a solar panel array to provide a voltage of 500V and the power rating determined in Task 1.

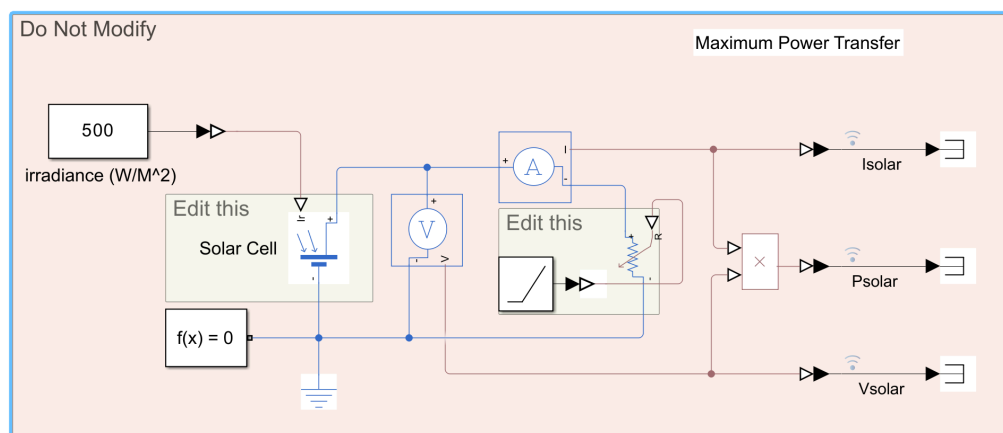


Image Task2_2

This ensured the array met the voltage and power requirements essential for MPPT integration in the next task.

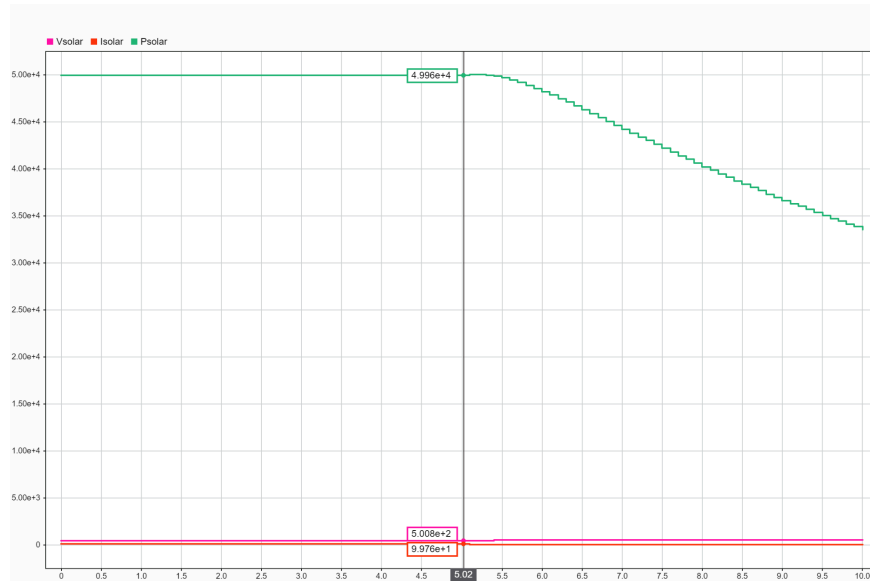


Image Task2_3

TASK 3 - DEVELOP MAXIMUM POWER POINT TRACKING CONTROL

The DC-DC boost converter parameters were tuned based on the operating voltage and current range of the solar panels.

The MPPT controller was initialized with the following configuration

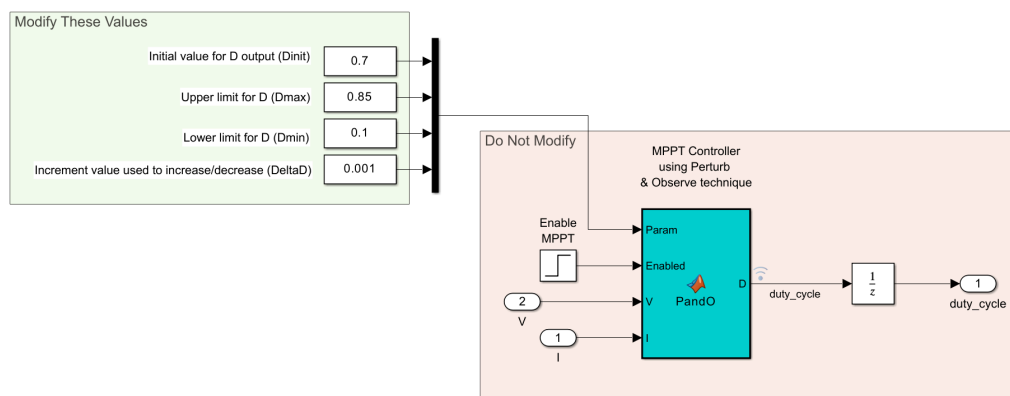


Image Task3_1

Simulation results showed that the MPPT system converged quickly to the optimal operating point, maintaining high tracking efficiency even with irradiance fluctuations and partial shading.

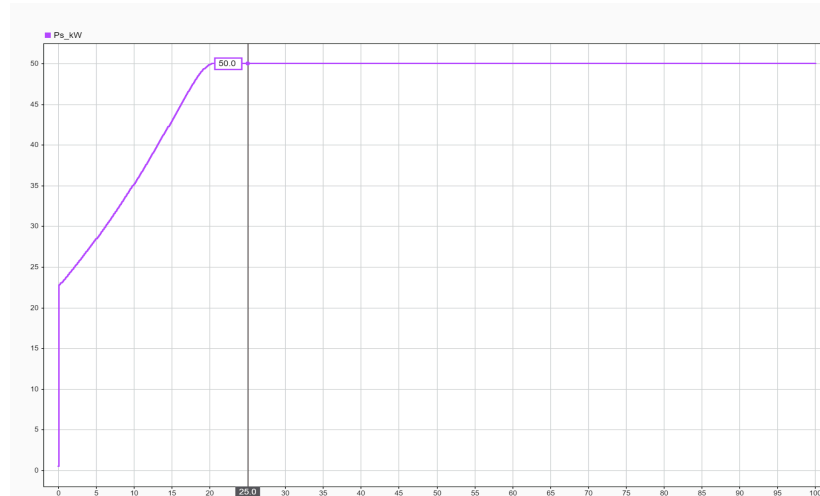


Image Task3_2

TASK 4 - DEVELOP BATTERY VOLTAGE CONTROL

A closed-loop voltage regulation strategy was developed to maintain the DC bus voltage within a critical operational band of 500 V to 520 V, as required for sensitive life-support equipment. A PI controller-based voltage regulation scheme was tuned. Modifications implemented included:

- Voltage Set Point (V_{ref}): Tuned to 510 V
- Controller Gains: $1e-8$

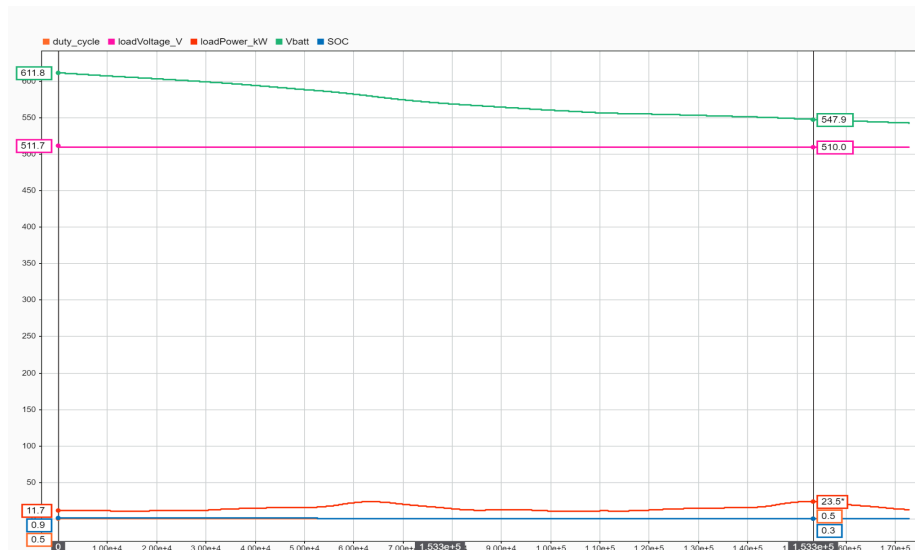


Image Task4_1

After tuning, the regulated voltage stayed well within the 500-510 V range across varying load conditions and charging/discharging cycles.

TASK 5 - INTEGRATE AND TEST THE FULL SYSTEM

The final task integrated all subsystems into the *Task5_FullSystem* model. The simulation ran for a full 7-sol duration with varying solar input. Key observations included:

- Uninterrupted Power Supply: Achieved for the entire duration.
- Voltage Range Maintained: Between 500 V - 520 V.
- SOC Limits Obeyed: Battery operated safely within 0%-100%.

This validated the design as technically sound and capable of sustaining life-support-critical systems in extreme extraterrestrial environments.

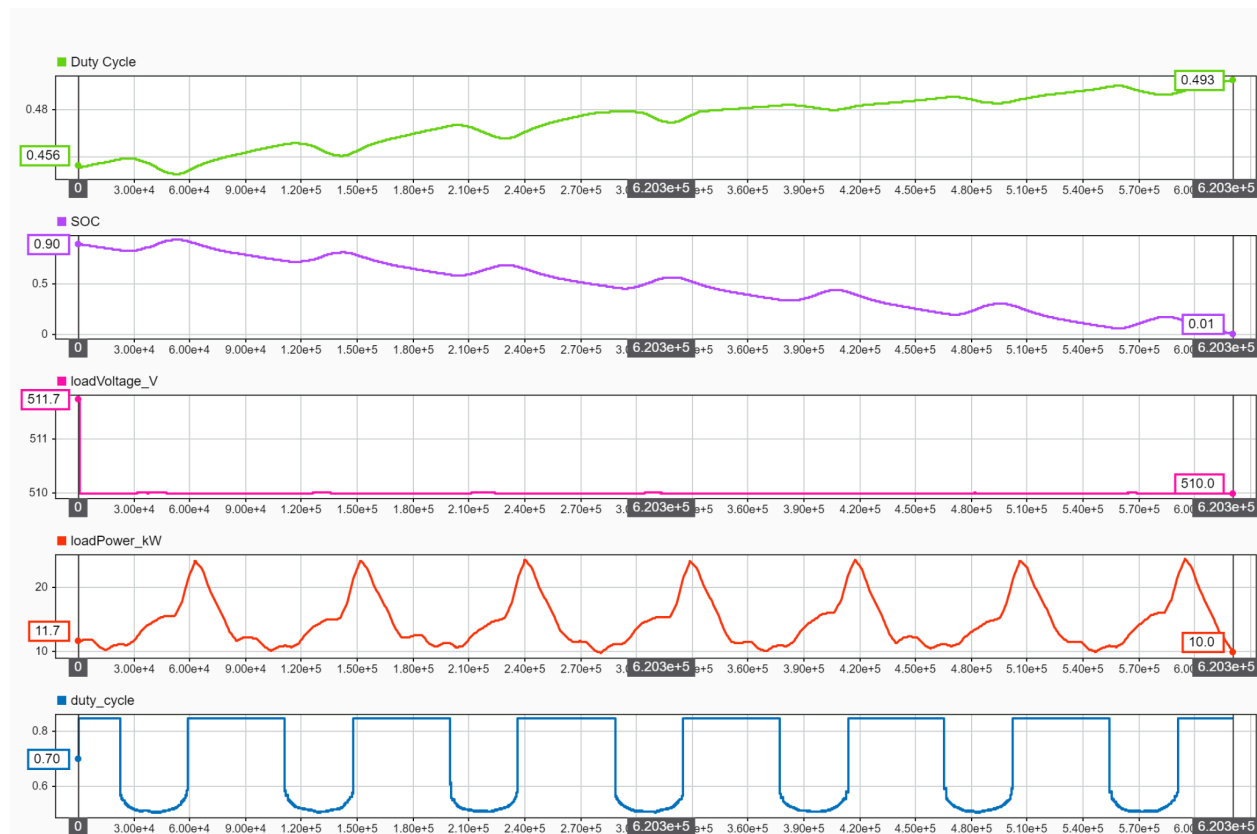


Image Task5_1

CONCLUSION

The "Microgrid on Mars" hackathon challenged us to design a reliable energy system for a Martian habitat. Through careful planning and testing, our team successfully built a working microgrid that provides steady power for a full 7-sol period.

Here are the key takeaways from our project:

- **Finding the Right Size:** We determined that a 50 kW solar array and a starting battery charge of 90% provided enough continuous energy without completely draining the battery.
- **Getting Maximum Power:** We extracted the maximum amount of power from our panels by finding the perfect load resistance (5.07 Ohms). We also arranged the panels (900 in series, 10 in parallel) to ensure we had the correct voltage.
- **Keeping it Stable:** We successfully tuned our control systems, including the MPPT algorithm and a voltage controller set to 510 V. This kept the power flowing smoothly and efficiently.
- **A Successful Final Test:** When we connected all the parts together, the complete system worked perfectly under simulated Martian conditions. It kept the voltage in the safe zone (500 V - 520 V) and managed the battery properly for the entire 7-day test.

Overall, this project showed us how important computer simulations are for designing space technology. We proved that with the right tuning and controls, it is possible to build a safe, continuous power supply for future missions to Mars.